

# Assignment 2 – SQL Queries

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## Case Study: Employee Performance Appraisal System

### 1. Aim

To design and implement a database for an Employee Performance Appraisal System and demonstrate the use of SQL DDL and DML statements including Create, Select, Update, Delete, operators, functions, and set operators.

### 2. Problem Statement (Case Study)

An organization wants to automate its Employee Performance Appraisal System to evaluate employees periodically. The system should store employee details, department information, appraisal records, and performance scores with ratings. The database should support employee evaluation, performance analysis, and department wise appraisal reports.

### 3. Database Design

Entities:

1. Department
2. Employee
3. Appraisal

Relationships:

- One department can have many employees
- One employee can have multiple appraisal records

### 4. Database Schema

Table 1: Department

dept\_id INT – Primary Key

dept\_name VARCHAR(50) – Department Name

location VARCHAR(50) – Department Location

Table 2: Employee

emp\_id INT – Primary Key  
emp\_name VARCHAR(50) – Employee Name  
designation VARCHAR(50) – Job Role  
salary DECIMAL(10,2) – Salary  
dept\_id INT – Foreign Key

Table 3: Appraisal

appraisal\_id INT – Primary Key  
emp\_id INT – Foreign Key  
appraisal\_year INT – Year  
performance\_score INT – Score (1-10)  
rating VARCHAR(20) – Rating

## 5. SQL Inputs and Outputs

### Create Department Table

```
CREATE TABLE Department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50) NOT NULL UNIQUE,  
    location VARCHAR(50)  
);
```

Output: Query OK, 0 rows affected

### Create Employee Table

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(50) NOT NULL,  
    designation VARCHAR(50),  
    salary DECIMAL(10,2) CHECK (salary > 0),  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES Department(dept_id)  
);
```

Output: Query OK, 0 rows affected

### Create Appraisal Table

```
CREATE TABLE Appraisal (  
    appraisal_id INT PRIMARY KEY,  
    emp_id INT,  
    appraisal_year INT CHECK (appraisal_year >= 2020),  
    performance_score INT CHECK (performance_score BETWEEN 1 AND  
10),  
    rating VARCHAR(20),  
    FOREIGN KEY (emp_id) REFERENCES Employee(emp_id)
```

```
);
```

Output: Query OK, 0 rows affected

### **Insert Department Data**

```
INSERT INTO Department VALUES
(1, 'Sales', 'Bangalore'),
(2, 'Engineering', 'Hyderabad'),
(3, 'Marketing', 'Chennai');
```

Output: 3 rows affected

### **Insert Employee Data**

```
INSERT INTO Employee VALUES
(201, 'Rohan Mehta', 'Sales Executive', 55000, 1),
(202, 'Priya Nair', 'Software Engineer', 65000, 2),
(203, 'Arjun Singh', 'Marketing Analyst', 48000, 3);
```

Output: 3 rows affected

### **Insert Appraisal Data**

```
INSERT INTO Appraisal VALUES
(11, 201, 2025, 8, 'Very Good'),
(12, 202, 2025, 9, 'Outstanding'),
(13, 203, 2025, 7, 'Good');
```

Output: 3 rows affected

### **Display Employee Table**

```
SELECT * FROM Employee;
```

| emp_id | emp_name    | designation       | salary   | dept_id |
|--------|-------------|-------------------|----------|---------|
| 201    | Rohan Mehta | Sales Executive   | 55000.00 | 1       |
| 202    | Priya Nair  | Software Engineer | 65000.00 | 2       |
| 203    | Arjun Singh | Marketing Analyst | 48000.00 | 3       |

### **Employees with Salary > 45000**

```
SELECT emp_name, salary FROM Employee WHERE salary > 45000;
```

| emp_name    | salary   |
|-------------|----------|
| Rohan Mehta | 55000.00 |
| Priya Nair  | 65000.00 |
| Arjun Singh | 48000.00 |

### **Employees in Department 2**

```
SELECT emp_name FROM Employee WHERE dept_id = 2;
```

```
emp_name
Priya Nair
```

### **Employee with Department Name**

```
SELECT e.emp_name, d.dept_name
FROM Employee e
JOIN Department d ON e.dept_id = d.dept_id;
```

```
emp_name      | dept_name
Rohan Mehta   | Sales
Priya Nair    | Engineering
Arjun Singh   | Marketing
```

### **Employees with Performance Score >= 8**

```
SELECT e.emp_name, a.performance_score
FROM Employee e
JOIN Appraisal a ON e.emp_id = a.emp_id
WHERE a.performance_score >= 8;
```

```
emp_name      | performance_score
Rohan Mehta    | 8
Priya Nair     | 9
```

### **Average Salary**

```
SELECT AVG(salary) AS average_salary FROM Employee;
```

```
average_salary
56000.000000
```

### **Employees from Dept 1 and 2 using UNION**

```
SELECT emp_name FROM Employee WHERE dept_id = 1
UNION
SELECT emp_name FROM Employee WHERE dept_id = 2;
```

```
emp_name
Rohan Mehta
Priya Nair
```

### **Department Table**

```
SELECT * FROM Department;
```

```
dept_id | dept_name      | location
```

|   |             |           |
|---|-------------|-----------|
| 1 | Sales       | Bangalore |
| 2 | Engineering | Hyderabad |
| 3 | Marketing   | Chennai   |

### Appraisal Table

```
SELECT * FROM Appraisal;
```

| appraisal_id | emp_id | appraisal_year | performance_score | rating      |
|--------------|--------|----------------|-------------------|-------------|
| 11           | 201    | 2025           | 8                 | Very Good   |
| 12           | 202    | 2025           | 9                 | Outstanding |
| 13           | 203    | 2025           | 7                 | Good        |

## 6. Output

- Successfully created tables with constraints
- Inserted records into Department, Employee, and Appraisal tables
- Executed Select, Update, and Delete queries
- Demonstrated SQL functions, operators, joins, and set operators

## 7. Conclusion

The Employee Performance Appraisal System database was successfully designed and implemented using SQL. DDL statements ensured data integrity using constraints, while DML queries demonstrated effective data manipulation and retrieval for appraisal analysis.